GGG-S-63C
August 18, 1976
SUPERSEDING
Fed. Spec. GGG-S-63b
April 17, 1963

FEDERAL SPECIFICATION

SAW, CIRCULAR, PORTABLE, PNEUMATIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable pneumatic circular saws, primarily intended for woodcutting, but also suitable, with the use of applicable blades (see 3.14), for cutting steel, stone, masonry, ceramics, plastics, and other ferrous and nonferrous materials.

1.1.1 Federal specification coverage. Federal specifications do not include all types and sizes of the commodities indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types. The saw shall be furnished in the following types in nominal 12-inch sizes, as specified (see 6.2).

Type A - Standard duty.
Type B - Heavy duty.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:
WW-C-633 - Couplings, Hose, Pneumatic, Universal Type.
GGG-B-440 - Blade, Circular Saw, Woodcutting.
PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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National Bureau of Standards (NBS) handbook:

H-28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

American National Standard Institute (ANSI) standard:

S5.1 - Test Code For The Measurement of Sound From Pneumatic Equipment.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

3. REQUIREMENTS

3.1 Qualification. Circular, portable, pneumatic saws furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Illustrations. The illustrations shown herein are descriptive, and not restrictive, and are not intended to preclude the purchase of saws otherwise conforming to this specification.

3.3 Materials. The materials used in the construction of all components shall be capable of completing the examinations and tests specified herein. All components shall be new and unused. Repaired or reconditioned components shall not be acceptable.

3.4 Screw threads. All threaded parts, including screws, bolts, and nuts, shall conform to the applicable requirements of Handbook H-28.

3.5 Identification markings. Each saw shall be marked in a permanent and legible manner showing at least the folloiwng: The manufacturers' name or registered trade mark, model number, size, and maximum no load speed.

3.6 Primary lubrication. Saws shall be provided with an oil reservoir. The oil reservoir shall be of sufficient capacity so that one filling will supply lubrication to moving parts in the path of the air stream for at least 8 hours of continuous operation. The oil shall be fed from the reservoir by an automatic device that functions through an action dependent upon the operation of the saw. The oil flow shall be regulated so that lubrication will be sufficient without excessive quantities of lubricant being discharged with the exhaust air.

3.6.1 Secondary lubrication. Conventional type lubrication fittings or openings or other suitable means shall be provided for introduction of sufficient lubricant for bearings, gears, and other moving parts not lubricated from the oil reservoir. The fittings and openings shall be such that lubrication is required only once in not less than 48 hours of actual operation. Where acceptable prelubricated sealed bearings are used, no external means of lubrication shall be required for these parts.

3.7 Air inlet connection. The air inlet connection shall consist of a 1/2 inch well-secured brass or ferrous fitting threaded with internal American Standard taper pipe threads (NPT) or internal Dryseal American Standard taper pipe threads (NPTG). The air inlet shall be fitted with an air strainer or screen (20 mesh or finer) which shall be effective in retaining solid particles in the compressed air supply. The strainer or screen shall be of such design that it can be easily removed for cleaning.

3.8 Description. The saws shall conform to the requirements shown in table I and be similar to figure 1. The components shall include among other necessary appurtenances, a rotary-vane air motor, an adjustable depth and tilting base, conveniently mounted handles, and a combination rip and crosscut saw blade (see 3.14). The saw shall operate with an air supply ranging from 80 to 100 pounds per square inch gage (lb/in^2g).

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Table I - Dimensional and performance requirements at 90 lb/in^{2}g air supply.

Nominal size	Class	Specified torque	Speed at specified torque	Air consumption (max)	Speed at no load (max)	Vert. depth cut	45° depth of cut	Weight without the blade
inches		foot-pounds	r/min (min)	scfm/hp	r/min	(min) Inches	(min) Inches	(max) Pounds
12	A	6.5	1300	32	3100	3-7/8	2-3/4	30
12	B	10	1800	35	2400			35

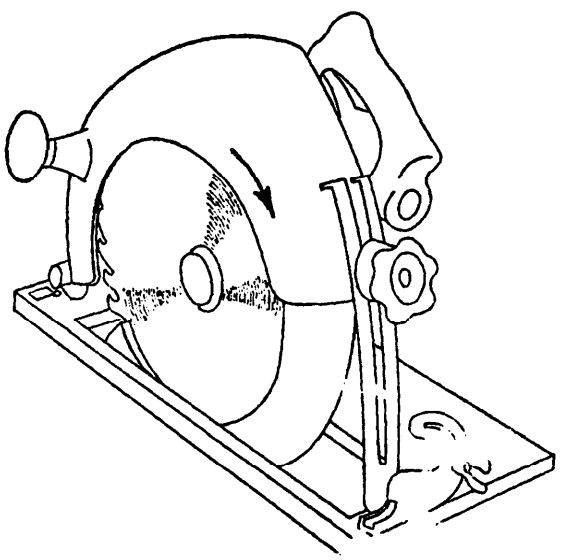


Figure 1 - Saw, circular, portable, pneumatic.

3.8.1 Sound pressure level. The sound pressure levels measured at an unobstructed distance of one meter, shall not exceed 95 decibels (dB) on the A-weighted scale in any prescribed microphone location (see 4.6.2).

3.9 Motor. The motor shall be the rotary-vane type having a single rotor with positive displacement. The design of the motor shall be such that a portion of the air flow is directed underneath the rotor vanes so that the vanes are held firmly against the cylinder wall at any rotating speed. A portion of the exhaust air shall be so directed that the dust and chips are blown from the cutting line of the blade. Provisions for limiting overspeed of the motor shall be incorporated.

3.10 Housing. The housing shall be strong and rugged to prevent indentation under service conditions. Through bolting shall be used wherever possible in aluminum-alloy shells. If threaded openings in an aluminum-alloy shell are necessary, a threaded depth (minimum) of one and one-half times the pitch diameter of the same thread or threaded ferrous inserts shall be provided on parts subject to load or wear. Nuts and bolts shall be secured by suitable locking devices. The side of the housing when viewed from the spindle end shall be clearly and permanently marked with an arrow to indicate the direction the blade is turning.

3.11 Handle. The saw shall have a closed pistol-grip type handle with an integral throttle valve. The throttle valve shall have a positive safety lock and shall be located within the closed pistol-grip handle to prevent accidental starting of the saw. An auxiliary handle or knob shall be provided and located in such a manner as to make the saw suitable for two-handed operation.

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3.12 Bearings. All rotating parts of the saw shall be provided with commercially available ball or roller bearings. Ball and roller bearings, which are primarily designed for radial loads, shall not be used to absorb axial thrust. Open-type bearings shall be suitably housed and adequately sealed to prevent leakage of lubricant and entrance of dirt and dust.

3.13 Arbor. The arbor shall be 0.750 or 1.000 $^{+0.000}_{-0.001}$ inches in diameter as specified (see 6.2). Means shall be provided to preclude the saw blade turning on the arbor in the event the stalled torque of the saw is reached.

3.13.1 Rotation. The saw-blade rotation shall be such that the material being cut will be held against the saw base by the cutting action of the saw-blade.

3.14 Saw blade. Unless otherwise specified (see 6.2), a 12-inch diameter rip and crosscut saw blade shall be furnished. The blade shall conform to type III, class I of GGG-B-440, have 18 style S teeth and arbor hole of size compatible with arbor specified 3.13.

3.15 Throttle. Saws shall be provided with a nonleaking manual throttle for admitting air supply, gradually regulating the speed of the tool, and automatically shutting off supply of compressed air when throttle is released.

3.16 Saw-blade guards. The saw shall be provided with guards above and below the base. The guards above and below the base shall cover the saw-blade to the depth of teeth. The guard below the base, except for the minimum arch of the saw-blade required to enter the work shall be automatic, self-closing telescoping type, which shall instantly return to covering the saw-blade when the saw is withdrawn from the work.

3.17 Base. The base shall support the saw against the material being cut, and shall be securely fastened to the saw housing. When specified (see 6.2), the saw shall be provided with means for mounting an adjustable rip-fence on either side to the base. The rip-fence furnished shall provide a minimum rip width of 6 inches.

3.17.1 Position indicator. Lines, grooves, or pointer shall be located at the leading edge of the base to clearly indicate exactly where the saw blade is going to enter the work when the blade setting is at zero angle and 45 degrees; lines, grooves, or pointer shall be visible from user's operating position.

3.17.2 Angle adjustment. Means shall be provided for adjusting the base from 0 to 45 degrees. A graduated scale shall be provided to clearly indicate angle setting in increments of not more than 5 degrees from 0 to 45 degree angle. Angle adjustment and fastening devices shall be accomplished without the use of tools.

3.17.3 Depth adjustment. The base shall be readily adjustable to control the depth of cut within the minimum vertical and 45 degree depth specified in table I. Depth adjustment and fastening devices shall be accomplished without the use of tools.

3.18 Performance. The saw shall satisfactorily complete the performance tests (see 4.6) and shall conform to the performance requirements shown in table I.

3.19 Accessories. One arbor nut wrench shall be included with each saw.

3.19.1 Hose assembly rubber. When specified (see 6.2), the saw shall be furnished with a 10-foot length of air line hose having a screw thread coupling on one end for connection to the air line inlet and a quick disconnect universal hose coupling conforming to type I of WW-C-633 on the other end.

3.19.2 Repair parts. The contractor shall furnish repair parts when specified (see 6.2). All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances in order that such parts may be completely interchangeable and may be replaced or adjusted without requiring modification.

3.20 Technical manual. One copy of a technical manual consisting of the manufacturer's standard commercial instructions and parts list, shall be furnished with each saw.

3.21 Carrying case. Each saw shall be provided with a carrying case. The carrying case shall be at least 22 gage steel and of rigid construction. Provision shall be made for securely retaining the cover closed. The case shall be of ample size and design to hold the saw and accessories in a fixed position. The interior and exterior surfaces of the carrying case shall have a durable coating of paint or enamel. The case shall be marked with the manufacturer's name or trademark and saw size.

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3.22 Workmanship. Workmanship of the saws shall be such that the tools shall be free of defects which would adversely affect the operation, efficiency, serviceability, and life expectancy of the tools. All components shall be new and machined to tolerances which insure ease of interchangeability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the contractor is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the examinations as specified in 4.5 and the tests specified in 4.6. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.3.1).

4.4 Quality conformance inspection.

4.4.1 Inspection lot. All saws of the same size offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105 at inspection level II. The acceptable quality level (AQL) shall be 1.5 percent defective. The sample unit shall be one completely fabricated saw with blades and carrying case.

4.4.3 Quality conformance tests. Quality conformance inspection shall be performed on each sample saw selected in accordance with 4.4.2. Quality conformance inspection shall consist of the examinations as specified in 4.5 and the tests specified in 4.6.1 and 4.6.3.

4.5 Examination.

4.5.1 Visual examination. Each sample unit selected in accordance with 4.4.2 shall be examined for conformance to the requirements specified in 3.3 thru 3.8, 3.9 thru 3.17.3 and 3.19 thru 3.22. Defects are listed in table II.

Table II - Classification of defects.

Category	Defects
Critical	
001	Moving parts improperly guarded.
Major	
101	Material nonconforming.
102	Saw not easily adjustable or parts not easily replaceable.
103	Lubricated parts enclosure nonconforming.
104	Similar parts not dimensionally and functionally interchangeable.
105	Air-inlet connection not as specified, damaged, or missing.
106	Air strainer missing or damaged, not easily removable.
107	Marking, nonconforming, illegible, incorrect or missing.

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Table III - Classification of defects (con).

Category	Defects
Major	
108	Handles not as specified or damaged, not removable when required.
109	Throttle nonconforming, damaged, inoperable or missing.
110	Spindle not as specified.
111	Bearings not as specified, broken, cracked, or pitted.
112	Lubrication of saw not as specified; evidence of oil leakage.
113	Blade rotation not indicated as specified.
Minor	
201	Marking, manufacturer's name or trademark missing, incorrect, illegible or not permanent.
202	Preservation, packaging and packing not as required.
203	Carrying case not as specified.

4.6 Performance tests.

4.6.1 Performance tests. All performance tests shall be based on operation of the saw with compressed air at 90 lb/in^{2}g measured as close to the tool as practicable. The saw shall be connected to the air supply by means of a 10 foot length 5/8 inch inside diameter hose. The saw shall be tested for air consumption, speed, and torque as specified in the applicable table.

4.6.2 Sound pressure level tests. The sound pressure level tests shall be conducted in accordance with the applicable provisions of S5.1 except as modified herein. The acceptable limits shall be as specified in 3.8.1.1.

4.6.2.1 Acoustical calibration. The entire instrumentation system shall be calibrated before each test series allowing a minimum instrument warm-up period of 5 minutes if transistorized and 1/2 hour if vacuum tube. Recalibration shall be performed if any instrument adjustment is necessary during progress of test.

4.6.2.2 Background sound level. The ambient noise level shall be at least 10 dB less than the measurement taken with the tool running in all tool and microphone orientations. In addition, the sound produced by the energy absorbing device used in the "on-load" test, must be at least 10 dB below the machines own sound output.

4.6.3 Air leakage test. Saws shall be tested for tightness. Saws shall be connected to a system consisting of a 0.25 cubic foot air receiver, valving as necessary and a 0 - 300 lb/in^2 range pressure gage. System shall be pressurized with air or nitrogen to 150 lb/in^2 with throttle in the closed position and isolated for a period of 15 minutes. At the end of 15 minutes, maximum pressure drop permitted shall be no more than 50 lb/in^2.

4.6.4 Service tests. Saws shall complete 150 hours of actual work under practical service conditions, to determine adequacy and maintainability of the tool. When test under practical service conditions is not feasible, a simulated test may be performed of equal duration by using a life testing machine. Tool shall be loaded to specified torque on a prony brake or other load applying device. Tool shall be cycled on for 20 seconds and off for 10 seconds.

4.6.4.1 Lubrication test. Lubrication system shall be checked to insure conformance with 3.6. This test may be performed concurrently with the service test. Lubricant reservoir shall be inspected periodically to determine adequacy of lubrication system.

4.6.4.2 After service test. Upon completion of the service test, the saw shall be disassembled and the various parts examined. Any saw having broken, cracked or deformed parts shall be rejected. Saws completing the service test shall be again subjected to the performance test of 4.6.2 and 4.6.3, and shall conform to the applicable requirements of this specification.

4.7 Inspection of preparation for delivery. The preservation, packaging, packing and marking shall be examined for conformance with the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Saws shall be preserved, packaged, packed and marked in accordance with PPP-P-40. The level of presefvation and packaging and level pf packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The portable pneumatic saws covered by this specification are intended for sawing to size of lumber, metal shapes and masonry products with the proper blade.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and class required (see 1.2.1) and arbor size required (see 3.13).
(c) Blade requirements, if other than specified (see 3.14).
(d) When an adjustable rip fence shall be furnished (see 3.17).
(e) Performance requirements, if other than specified (see 3.18).
(f) When a hose assembly shall be furnished (see 3.19.1).
(g) Repair parts to be furnished (see 3.19.2).
(h) Applicable levels of preservation and packaging and level of packing (see 5.1).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL GGG-S-63 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 This specification reflects the requirements of OSHA in effect as of the date of this specification.

Custodians:
Army - GL
Navy - SH
Air Force - 84

Review activities:
Army - GL
Navy - SH
Air Force - 84, 80

User activities:
Army - CE, ME
Navy - YD, MC

Preparing activity:
Navy - SH
(Project 5130-0160)

Civil Agency Coordinating Activity:

GSA-FSS

U. S. GOVERNMENT PRINTING OFFICE : 1977 - 241-233/1011

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DOCUMENT IDENTIFIER AND TITLE
GGG-S-63C, Saw, Circular, Portable, Pneumatic

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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